

UEC637: VLSI Testing and Verification

L T P Cr
3 0 2 4.0

Course Objective: In this course students will learn test economics, fault modelling, logic and fault simulation, ATPG concepts for combinational and sequential circuits. Students will also be able to write test bench for the complex VLSI design.

Introduction: Role of testing in VLSI design, Issues in test and verification of complex chips, VLSI test process and equipment, Test economics, Yield analysis and product quality.

Faults modelling and fault simulation: Physical faults and their modelling, Stuck-at faults, Bridging faults, Fault collapsing, Fault simulation, Deductive, Parallel and Concurrent fault simulation, Combinational and sequential SCOAP measures.

ATPG for combinational circuits: D-Algorithm, Boolean Difference, PODEM, Random, Exhaustive and Weighted Test Pattern Generation, Aliasing and its effect on Fault coverage.

ATPG for sequential circuits: ATPG for Single-Clock Synchronous Circuits, Time frame expansion Method, Simulation-Based Sequential Circuit ATPG.

Memory testing and BIST: Permanent, Intermittent and pattern sensitive faults, March test notion, Memory testing using march tests, PLA testing, Ad-Hoc DFT methods, Scan design, Partial scan design, Random logic for BIST, Memory BIST.

Verification: Design verification techniques based on simulation, Analytical and formal approaches, Functional verification, Timing verification, Formal verification, Basics of equivalence checking and Model checking, Hardware emulation.

Laboratory Work:

Familiarization with development of test benches using Verilog/System Verilog on Mentor/Cadence/Xilinx-ISE tools, Logic simulation, Logic level diagnosis, ATPG, development of Verification plan for the given design and writing testcases, computation of fault coverage/code coverage index

Text Books:

1. M. Bushnell and Vishwani Agrawal, Essentials of Electronic Testing for Digital, Memory and Mixed-Signal VLSI Circuits, Springer, ISBN 978-0792379911.
2. Chris Spear, System Verilog for Verification, Springer, ISBN 978-1-4614-0714-0

Reference Books:

1. M. Abramovici, M. Breuer, and A. Friedman, Digital System Testing and Testable Design, IEEE Press, 1994
2. Diraj K. Pradhan, "Fault Tolerant Computer System Design", Prentice Hall.
3. L. T. Wang, C. W. Wu, and X. Wen, VLSI Test Principles and Architectures, Morgan Kaufmann, 2006, ISBN-13: 978-0-12-370597-6, ISBN-10: 0-12-370597-5.
4. System-on-a-Chip Verification-Methodology and Techniques, P. Rashinkar, Paterson and L. Singh, Kluwer Academic Publishers, 2001
5. Janick Bergeron, "Writing test benches functional verification of HDL models" Kluwer Academic Publishers, New York, Boston, Dordrecht, London, Moscow, 2002.

Course Learning Outcomes:

The student will be able to

1. Acquire knowledge about fault modelling and collapsing.
2. Learn about various combinational and sequential automatic test pattern generation techniques.
3. Analyze different memory faults and its testing methods.
4. Develop the verification plan for the small to complex VLSI designs.
5. Develop test bench using HVL for testing and verification of VLSI designs.

Evaluation Scheme:

S.No.	Evaluation Elements	Weightage (%)
1.	MST+EST	70
2.	Sessional (May include Assignments//Quizzes)	30

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 114th meeting of the Senate held on March 11, 2024 and March 7, 2025 respectively.